

SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL
CHENNAI 600 063
COMPUTER SCIENCE -QUESTION BANK

OUTPUT PREDICTION QUESTIONS-BOARD BASED SET-1

1. What will be the output of the following code?

```
rec={"Name": "Python", "Age": "20"}
```

```
r= rec.copy()
```

```
print(id(r)==id(rec))
```

2. What will be the output of the following list operations?

```
data = [10,20,30,[40,50,60],[70,80]]
```

```
print(data[3]+data[-1])
```

```
print(data[-2][-2])
```

3. What will be the output of the following python program?

```
data = [10,20,30, 60,70]
```

```
data[3:3]=[40,50]
```

```
print(data)
```

```
data.pop(3)
```

```
print(data)
```

```
data.extend([10,20])
```

```
print(len(data))
```

```
import random
RUNS =[40,55,60,35,70,50]
BEGIN =random.randint(0,2)
END =random.randint(1,3)
for i in range(BEGIN,END+1):
    print(RUNS[i],end='#')
```

a) 60#35#

b) 60#35#70#50#

c) 35#70#50#

d) 40#55#60#

```
import random
MIN = 25
SCORE = 10
for i in range (1,5):
    Num = MIN + random.randint(0,SCORE)
    print(Num,end=":")
    SCORE-=1;
print()
```

a) 34:31:30:33:

b) 29:33:30:31:

c) 34:31:30:31:

d) 34:31:29:33:

```
Area=["NORTH","SOUTH","EAST","WEST"]
```

```
for l in range(3):
```

```
    ToGo=random.randint(0,1) + 1
```

```
    print(Area[ToGo],end=":")
```

```
print()
```

a) SOUTH : EAST : SOUTH :

b) NORTH : SOUTH : EAST :

c) SOUTH : EAST : WEST :

d) SOUTH : EAST : EAST :

```
import random
```

```
L=[10,7,21]
```

```
X=random.randint(1,2)
```

```
for i in range(X):
```

```
    Y=random.randint(1,X)
```

```
    print(L[Y],"S",end=" ")
```

(a) 10 \$ 7 \$

(b) 21 \$ 7 \$

(c) 21 \$ 10 \$

(d) 7 \$

```
def ChangelIt(Text,C):
    T=""
    for K in range(len(Text)):
        if Text[K]>='F' and Text[K]<='L':
            T=T+Text[K].lower();
        elif Text[K]=='E' or Text[K]=='e':
            T=T+C;
        elif K%2==0:
            T=T+Text[K].upper()
        else:
            T=T+T[K-1]
    print(T)
OldText="pOwERALone"
ChangelIt(OldText,"%")
```

```
def drawline(char='$',time=5):
    print(char*time)

drawline()
drawline('@',10)
drawline(65)
drawline(chr(65))
```

```
def makenew(mystr):
    newstr = ""
    count = 0
    for i in mystr:
        if count%2 ==0:
            newstr = newstr+i.lower()
        else:
            if i.islower():
                newstr = newstr+i.upper()
            else:
                newstr = newstr+i
        count +=1
    newstr = newstr+mystr[:3]
    print ("The new string is :", newstr)
makenew("cbs eEXAMs@2022")
```

6 Predict the output of the following code:

```
S="LOST"
L=[10,21,33,4]
D={}
for I in range(len(S)):
    if I%2==0:
        D[L.pop()]=S[I]
    else:
        D[L.pop()]=I+3
for K,V in D.items():
    print(K,V, sep="**")
```

```
def check(a):
    for i in range(len(a)):
        a[i]=a[i]+5
    return a
b={1,2,3,4}
c=check(b)
print(c)
```

```
L1=[20, 3, 67, 89, 22]
def updatelist(L):
    L[0]=L[1]+L[3]
    L[3:4]=[100,200,300]

print(L1)
updatelist(L1)
print(L1)
```

```

def Execute(M):
    if M%3==0:
        return M*3
    else:
        return M+10;
def Output(B=2):
    for T in range (0,B):
        print(Execute(T),"*",end="")
    print()
Output(4)
Output()
Output(3)

```

```

X = 100
def Change(P=10, Q=25):
    global X
    if P%6==0:
        X+=100
    else:
        X+=50
    Sum=P+Q+X
    print(P,'#',Q,'$',Sum)
Change()
Change(18,50)
Change(30,100)

```

CHAPTERWISE BOARD BASED -PROGRAM BASED QUESTIONS

FUNCTIONS

1. Write a python program that accepts a list of integers from user and creates a new list from the existing list containing all the numbers that have three or more digits. For Example: for existing list [10,100, 99,200,1000] the new list should be [100,200,1000]

2. Write a function SQUARE_LIST(L), where L is the list of elements passed as argument to the function. The function returns another list named 'SList' that stores the Squares of all Non-Zero Elements of L. For example: If L contains [9,4,0,11,0,6,0], the SList will be[81, 16, 121, 36]

3. Armstrong Number

Write a function:isArmstrong(n)that returns True if the number is an Armstrong number and False otherwise.

The program should accept a number and display an appropriate message.

Example: Enter number: 153 153 is an Armstrong number

Condition: Do not convert the number into a string.

4. Second Largest

Write a function:secondLargest(L) to find and return the second-largest element in a list without using sort() or max().

5. List Transformation

Write a function: transform(L) that creates and returns a new list containing:

- square of even numbers
- cube of odd numbers

For example: L = [2, 3, 4, 5] Output: [4, 27, 16, 125]

6. Number Analysis

Write a Python program using the following functions:

sumDigits(n)

reverseNumber(n)

isPalindrome(n)

The program should accept an integer and display:

- Sum of digits
- Reverse of the number
- Whether it is a palindrome

Example:

Enter number: 1221

Sum of digits = 6

Reverse = 1221

Palindrome = Yes

STACK

1. A list, NList, contains following record as elements: [City, Country, distance from Delhi]

Each of these records are nested together to form a nested list. Write the following user defined functions in Python to perform the specified operations on the stack named travel.

(a) Push_element(NList): It takes the nested list as an argument and pushes a list object containing name of the city and country, which are not in India and distance is less than 3500 km from Delhi.

(b) Pop_element(): It pops the objects from the stack and displays them. Also, the function should display "Stack Empty" when there are no elements in the stack.

For example: If the nested list contains the following data:

```
NList = [ ["Newyork", "USA", 11734], ["Naypyidaw", "Myanmar",3219],
["Dubai","UAE",2194], ["London","England",6693], ["Gangtok", "India",1580],
["Columbo", "Sri Lanka", 3405]]
```

The stack should contain:

```
["Naypyidaw", "Myanmar",3219], ["Dubai","UAE",2194], ["Columbo", "Sri Lanka", 3405]
```

The output should be:

```
["Columbo", "Sri Lanka", 3405]
```

```
["Dubai","UAE",2194]
```

```
["Naypyidaw", "Myanmar",3219] Stack
```

Empty

2. Coach Abhishek stores the races and participants in a dictionary. Write a program, with separate user defined functions to perform the

following operations:

(a) Push() Push the names of the participants of the dictionary onto a stack, where the distance is more than 100.

(b) PoP(): Pop and display the content of the stack. For example:

If the sample content of the dictionary is as follows:

Races = {100:'Varnika', 200:'Jugal', 400:'Kushal', 800:'Minisha'}} The output from the program should be: Minisha Kushal Jugal

Text files

1. Write a function COUNT() in Python to read from a text file 'rhym.txt' and display the

count of words in each line.

For Example: If the content of 'rhym.txt' is as follows

The function should display output as:

Line 1 : 3

Line 2 : 4

Line 3 : 2

<i>Jack and Jill</i> <i>Went up the hill</i> <i>To enjoy</i>
--

2. Write a function filter(oldfile, newfile) that copies all the lines of a text file "source.txt" onto "target.txt" except those lines which starts with "@" sign.
3. Write the definition of a Python function named LongLines() which reads the contents of a text file named 'LINES.TXT' and displays those lines from the file which have at least 10 words in it.
4. Write a program to read a text file "sample.txt" & display each word separated by a #.

BINARY FILES

1. A binary file restaurant.dat contains restaurant data structured as a dictionary: {Rno : [Rname, Rtype]}, where Rno is the Restaurant Number, Rname is the Restaurant Name, and Rtype is the Restaurant Type (e.g., 'Vegetarian', 'Non-Vegetarian', 'Multi-Cuisine').

Write a Python program with user-defined functions to perform the following:

- Search: VegNonVeg(Rtype) which accepts Rtype as a parameter and displays all records matching that type (e.g., 'Vegetarian').
- Update: UpdateRname(Rno, New_Name) to change the name of a restaurant for a given Rno.
- Delete: DeleteRestaurant(Rno) to delete a specific restaurant's dictionary entry from the file using its key (Rno).
- Count: CountRestaurants(Rtype) to count and return the total number of restaurants belonging to a specific Rtype.

2. A binary file named TEST.dat holds student test records. Each record is structured as a list: [TestId, Subject, MaxMarks, ScoredMarks].

Write a Python program to handle the data using these operations:

- Search & Average: DisplayAvgMarks(Sub) which accepts a subject name as an argument, searches for all student records belonging to that Subject, displays them, and calculates/prints the average ScoredMarks for that subject.
- Count: CountFailed(Sub, PassMark) to count how many students scored below the PassMark in a given Subject.
- Update: ModifyMarks(TestId, New_ScoredMarks) to update the marks scored by a student for a specific TestId.
- Delete: RemoveTestRecord(TestId) to delete a test record from TEST.dat using the TestId

CSV FILES

1. A CSV file employee.csv manages company payroll data. The header fields are: EmpID, Name, Department, Monthly_Salary.

Write a Python program to implement the following operations:

Create/Write: Write a function to accept employee details from the user and append 5 new records into employee.csv along with the header.

- Search: Display the details of all employees who belong to the "IT" department and earn a Monthly_Salary greater than 50000.
- Update: Write a function to update the Department of a specific employee by accepting their EmpID and the new department name.
- Delete: Write a function to remove an employee's row completely from the CSV file based on a given EmpID.
- Count: Count and display the total number of employees working in each unique department

2. A CSV file named delivery.csv holds shipment details for an e-commerce platform. The fields are: [OrderID, CustomerName, ItemName, OrderStatus, DeliveryCity]

Write a Python program to handle the data using these operations:

- Search: DisplayCityOrders(City) which accepts a city name as an argument, searches for all deliveries bound for that DeliveryCity, and displays them.
- Count: CountPendingOrders() to count and return the total number of orders where the OrderStatus is marked as "Pending".
- Update: MarkAsDelivered(OrderID) to search for an order by its OrderID and change its OrderStatus from "Shipped" or "Pending" to "Delivered".
- Delete: CancelOrder(OrderID) to delete an entire order record from delivery.csv if a customer cancels their purchase.

Python Output Prediction – Practice Questions

Topics: String • List • Tuple • Dictionary

Q1. Predict the output:

```
S = "DATA SCIENCE"  
print(S[::2])  
print(S[-2:2:-2])
```

Q2. Predict the output:

```
S = "PROGRAMMING"  
S=S.replace("M","*").replace("R","@")  
Print(S)  
print(S.count("G"))
```

Q3. Predict the output:

```
S = "MISSISSIPPI"  
print(S.find("ISS", 2))  
print(S.count("ISS"))
```

Q4. Predict the output:

```
S = " Python  Programming "  
L = S.strip().split()  
print("-".join(L))  
print(len(L[0]) + len(L[1]))
```

Q5. Predict the output:

```
S = "Python3.12"  
print(S.isalnum())  
print(S.replace(".", "").isalnum())  
print(S[:6].isalpha())
```

Q6. Predict the output:

```
S = "COMPUTER"  
L = list(S)  
L[1:4] = ["X","Y"]  
L.insert(-1, L.pop(0))  
print(L)  
print("".join(L))
```

Q7. Predict the output:

```
L = [4, 7, 2, 7, 9]  
x = L.pop(L.index(7))  
L.insert(0, x + L[-1])  
L.remove(2)  
print(L)
```

Q8. Predict the output:

```
L = [5, 1, 4, 2, 8]  
M = L.copy()  
L.sort()  
M.sort(reverse=True)  
print(L[1:4])  
print(M[-3:])
```

Q9. Predict the output:

```
L = [10, 20, 30]  
M = L  
N = L[:]  
M += [40]  
N.append(50)  
print(L)  
print(M)  
print(N)
```

Q10. Predict the output:

```
L = [2, 4, 6, 8, 10]  
for i in range(1, len(L)):  
    L[i] = L[i] - L[i-1]  
print(L)
```

Q11. Predict the output:

```
L = [1, 2, 3, 4, 5]
for i in range(len(L)-1, 0, -1):
    L[i] = L[i] + L[i-1]
print(L)
```

Q12. Predict the output:

```
L = [10, 20, 30, 40, 50]
A = L[1:4]
A.reverse()
A[0] = 99
print(L)
print(A)
```

Q13. Predict the output:

```
T = (10, 20, 30, 20, 40)
L = list(T)
L[T.index(20)] = 99
T = tuple(L)
print(T)
print(T.count(20))
```

Q14. Predict the output:

```
T = (1, 2, 3)
T += (4,)
T = T * 2
print(T[2:7])
print(T.count(3))
```

Q15. Predict the output:

```
T = (5, 10, 15, 20)

a, *b, c = T
print(a)
print(b)
print(c)
```

Q16. Predict the output:

```
D = {"A":1, "B":2, "C":3}
D.update({"B":20, "D":4})
x = D.popitem()
print(D)
print(x)
```

Q17. Predict the output:

```
D = {"A": [10,20], "B": [30,40]}
D["A"].append(D["B"].pop(0))
D["B"].insert(0, sum(D["A"]))
print(D)
```

Q18. Predict the output:

```
S = "BANANA"
D = {}
for ch in S:
    D[ch] = D.get(ch, 0) + 1
print(D)
print(list(D.keys())[::2])
```

Q19. Predict the output:

```
D = {"Maths":90, "CS":95,
      "English":85}
for k in list(D):
    if D[k] >= 90:
        D[k] += 5
    else:
        D[k] -= 5
print(D)
print(sum(D.values()))
```

Q20. Predict the output:

```
L = [3, 1, 4, 1, 5]
D = {}
for x in L:
    D[x] = D.get(x, 0) + 1
print(D)
print(sorted(D.keys()))
```

REVISION 2

Q21. Predict the output:

```
L = [10, 20, 30, 40]
for i in range(len(L)):
    if i % 2 == 0:
        L[i] += 5
    else:
        L[i] //= 2
print(L)
```

Q22. Predict the output:

```
D = {"A":10, "B":20, "C":30}
keys = list(D.keys())
for k in keys[::-1]:
    D[k] = D[k] + D[keys[0]]
print(D)
```

Q23. Predict the output:

```
L = [1,2,3,4]
A = L
B = L.copy()
A[1:3] = [20,30,40]
B[0] = 99
print(L)
print(B)
```

Q24. Predict the output:

```
D = {"A":10, "B":20}
for k in D:
    D[k] = D[k] // 10
D["C"] = D.get("A") + D.get("B")
print(D)
```

Q25. Predict the output:

```
S = "LEVEL"
print(S == S[::-1])
print(S.replace("E","3").replace("L","1"))
print(S.count(S[0]))
print(S)
```

Q26. Predict the output:

```
D = {"P": [1,2], "Q": [3,4]}
for k in D:
    D[k] = D[k] + [sum(D[k])]
print(D)
```

Q28. Predict the output:

```
S = "DATA SCIENCE"
L = S.split()
for i in range(len(L)):
    L[i] = L[i][::-1]
print(L)
print(" ".join(L))
```

Q30. Predict the output:

```
D = {}
for x in [2,3,2,4,3,2]:
    D[x] = D.get(x,0) + x
print(D)
print(max(D.values()))
```

Q31. Predict the output:

```
T = (10,20,30,40,50)
L = list(T[1::2])
L.insert(0, T[-1])
print(L)
print(tuple(L[::-1]))
```

Q32. Predict the output:

```
S = "abracadabra"
D = {}
for ch in S:
    if ch in "aeiou":
        D[ch] = D.get(ch,0) + 1
print(D)
```

Q34. Predict the output:

```
L = [10, 20, 30, 40, 50]
print(L.pop(2))
L.insert(2, L[-1])
L.remove(20)
print(L)
```

Q35. Predict the output:

```
S = "COMPUTER"
T = tuple(S[1:-1])
D = {ch:T.count(ch) for ch in T}
print(T)
print(D)
```

Q36. Predict the output:

```
D = {"A":10, "B":20, "C":30}
L = list(D.values())
L[1] = L[1] + D["A"]
D["D"] = sum(L)
print(L)
print(D)
```

print("Q37")

```
d = {"x": 5, "y": 10}
v1 = d.get("z", [])
v1.append(1)
v2 = d.get("z", [])
print(v1, v2, d)
```

print("Q38")

```
d1 = {"a": 1, "b": 2}
d2 = {"b": 3, "c": 4}
d1.update(b=10, d=5)
d1.update(d2)
```

```
print(d1)
```

print("Q39")

```
d = {"first": 1, "second": 2}
d["third"] = 3
d.pop("second")
d["fourth"] = 4
item1 = d.popitem()
item2 = d.popitem()
print(item1, item2, d)
```

print("Q40")

```
d = {"a": 1}
pairs = [("b", 2), ("a", 3)]
d.update(pairs)
print(d)
```

print("Q41")

```
d = {"a": 1, "b": 2, "c": 3}
d.pop("a")
```

```
d["a"] = 4
print(d.popitem())
```

print("Q42")

```
d = {"a": None, "b": 0}
print(d.get("a", "Missing"), d.get("c",
"Missing"))
print(d.pop("a", "Missing"), d.pop("a",
"Missing"))
```

print("Q43")

```
d = {"score": 50}
d.update({"score": 60}, score=70)
print(d)
```

print("Q44")

```
d = {"k1": "v1", "k2": "v2", "k3": "v3"}
k, v = d.popitem()
val = d.pop(k, "fallback")
print(k, v, val)
```

print("Q45")

```
d = {"count": 10}
d.update(total=100, count=20)
res_get = d.get("total", d.pop("count"))
res_pop = d.pop("count", d.get("total"))
last = d.popitem()
print(res_get, res_pop, last, d)
```

print("Q46")

```
d = {"alpha": 100, "beta": 200}
r1 = d.pop("alpha", d.get("beta"))
r2 = d.pop("alpha", d.get("beta"))
r3 = d.pop("beta", d.get("beta", 999))

print(r1, r2, r3, d)
```

Working with Functions – Advanced Board-Level Practice

Q1. Predict the output:

```
def change(a, b):  
    a = a + b  
    b = a - b  
    return a, b  
x = 10  
y = 5  
x, y = change(x, y)  
print(x, y)
```

Q2. Predict the output:

```
x = 20  
def fun():  
    x = 10  
    print(x, end=" ")  
    return x + 5  
print(x, end=" ")  
x = fun()  
print(x)
```

Q3. Predict the output:

```
def test(a, b=5, c=10):  
    return a + b * c  
  
print(test(2))  
print(test(2, 3))  
print(test(2, 3, 4))
```

Q4. Predict the output:

```
def modify(L):  
    L = L + [40]  
    L[0] = 100  
    return L
```

```
A = [10, 20, 30]
B = modify(A)
```

```
print(A)
print(B)
```

Q5. Predict the output:

```
def modify(L):
    L.append(40)
    L[0] = 100
```

```
A = [10, 20, 30]
modify(A)
print(A)
```

Q6. Predict the output:

```
def fun(n):
    if n <= 1:
        return 1
    return n * fun(n - 1)
print(fun(5))
```

Q7. Predict the output:

```
def f(n):
    if n == 0:
        return
    print(n, end=" ")
    f(n - 2)
```

```
f(7)
```

Q8. Predict the output:

```
def fun(a):
    a = a + 10
    print(a, end=" ")
```

```
x = 5
fun(x)
print(x)
```

Q9. Predict the output:

```
def process(a, b):
    a.append(b)
    b = b + 10
    return a, b
```

```
L = [1, 2]
x = 5
```

```
L, x = process(L, x)
```

```
print(L)
print(x)
```

Q10. Predict the output:

```
def calculate(a, b):
    return a + b, a * b
```

```
x = calculate(4, 5)
```

```
print(x)
print(x[0] + x[1])
```

Q11. Predict the output:

```
x = 5
```

```
def fun():
    global x
    x = x + 10
    print(x)
```

```
fun()
print(x)
```

CBSE cluster -2024

1. State True or False?

The order of exception handling in Python is try, followed by except, and then finally.

2. From the execution of the statements $n=300$, $m=1$, how many objects and references will be created?

- a) two objects 2 references b) one object 2 references
c) two objects 1 reference d) one object 1 reference

3. What gets printed after the execution of the following?

```
def x():
```

```
    pass
```

```
print(type(x))
```

- a) <class'NoneType'> b) <class'str'> c) <class'function'> d) Error

4. Given the tuple: $t=(10,20,30,50)$

Which of the following statements results in an error?

- a) `print(t[0])` b) `t.insert(2,3)` c) `print(t[1:2])` d) `print(len(t))`

5. What will be the output for the following Python statement?

```
d=("Amit","Sumit","Ashish","Sumanta")
```

```
print(d.index("Sumit") + 4//3 ** 2)
```

- a) 2 b) 4 c) 1 d) 3

6. Which of the following is not a keyword?

- a) None b) pickle c) global d) return

7. Which of the following will result in True?

- a) `math.pow(2,-3)+ math.sqrt(121) > 12`
b) `math.ceil(19.22)+ math.floor(15.99) > 34`
c) `math.sqrt(100// 10 - 6) == 5` d) Both a and b

8. For a given function header as follows: `def CALC(x, y=20):`

Which of the following function calls will give an error?

- a) `CALC(15,25)` b) `CALC(x=15,y=25)` c) `CALC(y=25)` d) `CALC(x=25)`

9. What will be the output of the following code?

```
Txt = "how are you, Sunil?"
```

```
X = Txt.find(",")
```

```
Y = Txt.find("?")
```

```
print(Txt[X+2:Y])
```

10. Suppose S is assigned as follows:

S = "Cluster"

All of the following expressions produce the same result except one.

Which one?

- a) S[:6] b) S[:-1][:-6] c) S[:1][:2] + S[len(S)-1] d) S[0] + S[-1]

11. Identify the output of the following code:

L = [6,4,2,9,7]

L[3:] = "100"

print(L)

- a) [6,4,2,9,7,100] b) [6,4,2,9,7,'1','0','0']
c) [6,4,2,1,0,0] d) [6,4,2,'1','0','0']

12.

- a) Both A and R are correct and R is the correct explanation of A.
b) Both A and R are correct and R is not the correct explanation of A.
c) A is True, R is False.
d) A is False, R is True.

Assertion: The tell() method returns an integer that specifies the current position of the file object in the file.

Reason: Offset can have any of the three values 0, 1 and 2.

13. If Data = { "students": [{"first name": "Nikhil", "last name": "Roy"}, {"first name": "Maya", "last name": "Devi"}, {"first name": "Suraj", "last name": "Kumar"}], "teacher": [{"first name": "Nisha", "last name": "Menon"}, {"first name": "Rakesh", "last name": "Varma"}, {"first name": "Amber", "last name": "Calico"}] }

- a) Write a Python statement to get a list of all student first names from the given dictionary.
b) Write a Python statement to count the total number of people (students and teachers) in the given dictionary.
c) How is mutable object different from immutable object in Python?
d) What is the output ? min(max(False,-3,-4),2,7)

14. Identify the proper built-in function/method for the following:

- a) To return a value which is less than or equal to a specific expression or value.
b) To convert upper case letter to lower case and vice-versa of the given string.

15. Sameer has written a Python program to count the occurrences of elements of a list using a dictionary. But the code has some syntax and logical errors. Re-write the code after removing all such errors and underline your correction.

```
L = [1,2,1,1,2,1,2,3,5,4,1,2,5,4,4,5]
```

```
D = {}
```

```
for x in L:
```

```
    if x not in D:
```

```
        D[x] += 1
```

```
    Else:
```

```
D[x] = 1
```

```
print(d)
```

16. A) Write the output of the following code.

```
def fun(k, X=[]):
```

```
    X.append(X.append(k))
```

```
    return X
```

```
for k in range(1,3):
```

```
    Y = fun(k)
```

```
    print(Y)
```

B)

```
def sample(l,k=[]):
```

```
    k.append(l)
```

```
    return k
```

```
print(sample(1))
```

```
print(sample(2))
```

```
print(sample(3,[]))
```

```
print(sample(4))
```

17. a) Write a function `countwords()` that reads a text file "story.txt" and displays only unique words from the file.

Example:

If the file story.txt contains the following:

"he is a good boy and he is also good by behaviour"

Then the output should be:

a boy and also good by behaviour

18. Write a function that reads a text file "diary.txt" and displays the list of words that start with a lowercase vowel and list of words that start with an uppercase vowel separately.

Example:

If the file diary.txt contains:

A quick black Elephant enters into a jungle.

Then the output should be:

List of words starting with lowercase vowels ["enters", "into", "a"]

List of words starting with uppercase vowels ["A", "Elephant"]

19. a) Lakshi has created a dictionary "Garment" storing data of garments items. Write a program for the following:

`Pushitems()` – the dictionary should be passed as parameter. The

function should check the details in the dictionary and push only the garment into a stack whose price is above 5000.

Popitem() – function to check stack emptiness; if it is not empty remove an item from the stack and display the remaining stack.

The structure of the dictionary is given below:

Garment = {'1':['shirt',6500], '2':['trouser',2000], '3':['coat',15000]}

20. John is assigned to write a program that inputs many numbers and stores only the perfect numbers into a stack named perfectstack[]. He has defined

the following functions for the purpose. Help him to complete the task.

Push-perfect() – to push the perfect numbers into the stack.

Pop-perfect() – to check the stack emptiness and pop an element from the stack if it is not empty and display the remaining stack.

21. Assume a CSV file marks.csv containing records of students as per the following structure:

name,class, marks separated by colon.

Write a Python program to perform the following using functions as name, class and marks separated by colon.

- Copycsv() : function to copy only the name and marks to another CSV file "class.csv".
- Copyslected() : function to copy only the records of students of class 2 to another CSV file "class2.csv".

For example:

Marks.csv

Nikhil:2:85

Maya:1:92

Suraj:2:78

Riya:3:88

class.csv

Nikhil,85

Maya,92

Suraj,78

Riya,88

class2.csv

Nikhil,2,85

Suraj,2,78

22. a) Which file can be opened with Notepad as well as MS Excel?

b) A binary file "Garment.dat" exists containing records of different types of garments as per the following structure: GID, Type, Gender, Cost

Write a Python program to add and search records of garments from the binary file using the following functions:

□ Addgarment() – function to input details of garments and store them to the file "Garment.dat" if the garment type is "cotton" or "silk".

□ Showgarment() – to read and display the records of "Silk" type and also display the count of the records displayed.